RAG Learning

2.1 Sampling

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs p21–p25

What this topic covers. The distinction between a population and a sample, the six named sampling techniques, how to select and evaluate a method for a given problem, sources of bias, and the role of the large data set.

1. Population and sample

Term	Meaning
Population	The entire group being studied.
Sample	A subset of the population, used to draw conclusions about it.
Census	Data collected from every member of the population.
Sampling frame	A list of all population members, from which the sample is drawn.
Parameter	A numerical property of the *population*.
Statistic	A numerical property calculated from a *sample*.

A sample statistic is used as an **estimate** of the corresponding population parameter — the sample mean estimating the population mean, the sample variance estimating the population variance (spec ref p22).

Census or sample?

	Census	Sample
Accuracy	Complete, no sampling error	Subject to sampling variation
Cost and time	High	Low
Practicality	Impossible if testing is destructive	Always possible

The destructive-testing point is the one that reliably earns a mark: you cannot census the lifetime of light bulbs, because testing every bulb destroys the entire stock.

2. Sampling techniques

MEI names six (p23, p24). Any other technique will be explained in the question.

Technique	Method	Strength	Weakness
Simple random	Every sample of the required size is equally likely	Unbiased	Needs a complete sampling frame
Systematic	Every kth member of an ordered list	Simple, spreads across the list	Fails if the list has a matching periodic pattern
Stratified	Sample each subgroup in proportion to its size	Represents every subgroup	Subgroup sizes must be known
Quota	Select until a set number in each category is filled	No sampling frame needed	Selection within categories is not random
Cluster	Divide into groups, sample whole groups	Cheap over a wide area	Clusters may not represent the population
Self-selected	Participants volunteer	Easy and cheap	Strongly biased towards those with strong views

Opportunity sampling — taking whoever is conveniently available — is also named. It is quick but unlikely to represent the population.

Quota and stratified sampling are the pair most often confused. Both fix the numbers from each category; the difference is that stratified sampling then selects **at random** within each stratum, while quota sampling does not.

Stratified sample calculations

Worked example. A school has 480 students in Year 12 and 320 in Year 13. Describe a stratified sample of 120.

Total 800, so the sampling fraction is $\frac{120}{800} = 0.15$.

$$\text{Year 12: } 480 \times 0.15 = 72, \qquad \text{Year 13: } 320 \times 0.15 = 48$$

Then select 72 and 48 students **at random** within each year group.

Two marks are commonly dropped here: not saying that selection within each stratum is random, and rounding each stratum independently so the totals no longer add to the required sample size.

3. Bias

A sampling method is **biased** if it systematically favours certain outcomes, so the sample is not representative.

Source	How it arises
Incomplete sampling frame	Some members can never be selected — a telephone directory omits those without listed landlines.
Non-response	Those who do not reply differ systematically from those who do.

Self-selection	Volunteers tend to hold stronger views than the population.
Time or place of sampling	Tuesday-morning shoppers differ from the full customer base.
Systematic periodicity	Sampling every 12th item when a fault recurs every 12th item.
Leading questions	The wording pushes respondents towards an answer.

Worked example. A supermarket surveys shoppers on a Tuesday morning to estimate mean weekly spending. Criticise the method.
Only shoppers available on a weekday morning can be selected, which under-represents people in full-time work — a group whose spending patterns may differ systematically. Non-response from those in a hurry compounds it. A better design samples across several days, times of day and stores.

When asked to criticise, name the source of bias, say **who is over- or under-represented**, and explain why that distorts the estimate. A bare statement that the sample "is biased" is not enough.

4. Sampling variability

Different samples from the same population give different results (p22). Two researchers taking independent random samples and obtaining different means have not made an error — that variation is inherent.

Larger samples give more reliable estimates: the standard deviation of the sample mean is $\frac{\sigma}{\sqrt{n}}$, so **quadrupling** the sample size halves it. Note that increasing the sample size reduces *variability* but does nothing about *bias* — a badly designed large sample is still badly designed.

5. The large data set

A large data set (LDS) is issued in advance and is intended to be explored in class using a spreadsheet or statistical software, so that its contexts become familiar.

- Questions in the Statistics component assume familiarity with those contexts.
- In the examination there is **no printout**, though selected data or summary statistics may be given in the paper.
- Familiarity is intended to give a material advantage, allowing questions that would be too demanding in an unfamiliar setting.
- From September 2026, all students on a two-year AS/A Level programme use a **single** large data set, replacing the earlier rotation of three.

The point of the LDS is *context*, not memorised numbers. Know what the variables mean, what units they use, how they were collected, and what would count as a plausible value — not individual entries.

6. Common pitfalls

- Confusing a parameter (population) with a statistic (sample).
- Describing stratified sampling without mentioning random selection within strata.
- Rounding stratum sizes so they no longer sum to the sample size.
- Confusing quota with stratified sampling.
- Saying a sample "is biased" without identifying who is misrepresented.
- Claiming a larger sample removes bias — it reduces variability only.
- Treating differences between two random samples as an error.